

**Paper : DSE-A-3  
(Computer Graphics)****Full Marks : 50**Answer **question no. 1** and **any four** questions from the rest.

1. Answer **any five** questions : 2×5
- (a) What do you mean by a pixel?
  - (b) Differentiate between physical and logical units of graphics devices.
  - (c) Define : pixmap and bitmap.
  - (d) What is aspect ratio?
  - (e) Write down the advantages of LCD.
  - (f) Define clipping operation.
  - (g) What do you mean by computer art?
  - (h) Define projection in graphics.
2. (a) Differentiate between Raster scan and Vector scan.
- (b) Can a color monitor display a black and white image? Explain.
- (c) What do you mean by resolution of an image? Are image resolution and screen resolution same? Explain properly. 3+3+(2+2)
3. (a) Discuss about different types of scaling operations.
- (b) What is shearing? Explain with proper example.
- (c) Mention the importance of inverse transformation. 4+4+2
4. (a) A triangle is defined by P(2, 2), Q(4, 2), R(4, 4). Find the transformed triangle co-ordinates after 60° rotation about origin in clockwise direction.
- (b) Find the transformation matrix for reflecting a point P(5,10) about the line  $y = -x$ . 5+5
5. (a) Write the Bresenham's line drawing algorithm.
- (b) Calculate the points between the starting co-ordinates (9, 18) and ending co-ordinates (14, 22) using Bresenham's line drawing algorithm. 5+5
6. (a) Discuss Cohen-Sutherland line clipping algorithm.
- (b) Clip a line A(-1, 5), B(3, 8) using the Cohen-Sutherland algorithm with window co-ordinates (-1, 1) and (2, 6). 5+5

**Please Turn Over****(0785+0786+0787)**

7. (a) Show that the composition of two rotations is additive by concatenating the matrix representations  $R(\theta_1)$  and  $R(\theta_2)$  to obtain  $R(\theta_1) \cdot R(\theta_2) = R(\theta_1 + \theta_2)$ .  
(b) A polygon has 4 vertices located at A(20, 10), B(60, 10), C(60, 30), D(20, 30). Indicate a transformation matrix to double the size of the polygon with point A located at the same place. 5+5
8. (a) Briefly describe different types of perspective projections.  
(b) Write a short note on computer animation.  
(c) Briefly discuss about morphing. 4+3+3